

1 Algorithmic Analysis

Asymptotic Notations

Big-O Notation

The upper bound for runtime/space.

$$T(n) \in O(f(n)) \iff$$

$$\exists c, n_0 (c > 0 \wedge n_0 > 0 \wedge \forall n > n_0 (T(n) \leq cf(n)))$$

Big-Ω notation

The lower bound for runtime/space.

$$T(n) \in \Omega(f(n)) \iff$$

$$\exists c, n_0 (c > 0 \wedge n_0 > 0 \wedge \forall n > n_0 (T(n) \geq cf(n)))$$

$$\text{Note: } f(n) \in O(g(n)) \iff g(n) \in \Omega(f(n))$$

Big-Θ notation

The bound for runtime/space.

$$T(n) \in \Theta(f(n)) \iff$$

$$T(n) \in \Omega(f(n)) \wedge T(n) \in O(f(n))$$

Comparison

$$O(1) < O(\log(\log(n))) < O(\log(n)) < O(\log^k n) <$$

$$O(n) < O(n^k) < O(2^n) < O(2^{2n}) < O(n!)$$

Akra-Bazzi Theorem

$$T(n) = g(n) + \sum_{i=1}^k a_i T(b_i n + h_i(n)), \text{ with}$$

- Sufficient base cases
- $a_i > 0, 0 < b_i < 1$
- $|g'(x)| = O(x^c), |h_i(x)| = O(\frac{x}{\log^2 x})$

$$\text{Solve for } p: \sum_{i=1}^k a_i b_i^p = 1$$

$$T(n) = \Theta(n^p (1 + \int_1^n \frac{g(u)}{u^{p+1}} du))$$

Master Theorem

$$T(n) = aT(n-b) + O(n^k), \text{ where } a, b > 0, k \geq 0$$

$$T(n) = \begin{cases} O(n^k) & a < 1 \\ O(n^{k+1}) & a = 1 \\ O(a^{n/b} n^k) & a > 1 \end{cases}$$

$$T(n) = aT(n/b) + \Theta(n^k \log^p n), \text{ where}$$

$$a \geq 1, b > 1, k \geq 0, p \in \mathbb{R}. \text{ For } c = \log_b a,$$

$$T(n) = \begin{cases} \Theta(n^k \log^p n) & c < k, p \geq 0 \\ \Theta(n^k) & c < k, p < 0 \\ \Theta(n^c \log^{p+1} n) & c = k, p > -1 \\ \Theta(n^c \log \log n) & c = k, p = -1 \\ \Theta(n^c) & c = k, p < -1 \\ \Theta(n^c) & c > k \end{cases}$$

Linear Recurrence

$$T(n) + c_1 T(n-1) + \dots + c_k T(n-k) + f(n)$$

For $f(n) = 0$:

$$x_n = c_1 x_{n-1} + \dots + c_k x_{n-k}$$

For t distinct roots of multiplicity m_i ,

$$T(n) = (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n +$$

$$\vdots$$

$$(\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n$$

For $f(n) \neq 0$, find solution a_h for $f(n) = 0$:

$$T(n) = a_p + a_h$$

Guess a_p similar to $f(n)$: Polynomial of degree n , An^n , Ac^n

2 Searching

Binary Search

Precondition: Array is sorted, and of size n

Postcondition: If element is in arr, A[begin]=key

Invariants: Key is within A[begin..end]

(end-begin) $\leq \frac{n}{2^k}$ after the k -th iteration

int search(A, key, n)

begin = 0

end = n-1

while begin < end do:

mid = begin + (end - begin)/2

if key <= A[mid] then

end = mid

else begin = mid + 1

return (A[begin]==key) ? begin : -1

Runtime: $O(\log n)$

Peakfinding

Precondition: Array of size n

Postcondition: $A[i-1] \leq A[i] \wedge A[i+1] \leq A[i]$

Invariants: $\exists$ a peak in A[begin..end]

Every peak in A[begin..end] is a peak in A[0..n-1]

Recurse in the right $\Rightarrow$ Peak in the right half is a peak in the array.

int findPeak(A, n)

if A[n/2] is a peak then return n/2

else if A[n/2 + 1] > A[n/2] then

return findPeak(A[n/2 + 1..n], n/2)

else if A[n/2 - 1] > A[n/2] then

return findPeak(A[1..n/2 - 1], n/2)

Runtime: $O(\log n)$ **Steep peakfinding:** $O(n)$

Quickselect

Precondition: Array of size n **Postcondition:** Return the k -th smallest element. **Invariants:** The k -th element is on the side that we recurse on.

Runtime: $O(n)$

3 Sorting

Problem: Array A[1..n] of elements $\Rightarrow$ Permutation B[1..n]

st. $B[1] \leq B[2] \leq \dots \leq B[n]$ *sorted - no diff*

Algo	Worst case	Best case	Space
Bubble	$O(n^2)$	$O(n)$	$O(1)$
Selection	$O(n^2)$	$\Omega(n^2)$	$O(1)$
Insertion	$O(n^2)$	$O(n)$	$O(1)$
Merge	$O(n \log n)$	$O(n \log n)$	$O(n)$
Quick	$O(n^2)$	$O(n \log n)$	$O(\log n)$
Heap	$O(n \log n)$	$O(n \log n)$	$O(1)$

Invariants

Bubble: Largest j items sorted in the final j positions after the j -th iteration Bubble sort is stable.

Selection: Smallest j items sorted in the first j positions after the j -th iteration Selection sort is non-stable.

Insertion: First j positions are sorted after the j -th iteration Insertion sort is stable.

Merge: tmp[j] is the minimum of all remaining elements after the j -th iteration Mergesort is stable.

Quick: B is partitioned around pivot

A[high] > pivot at the end of each iter

void quickSort(arr, n)

if (n == 1) then return;

else

p = partition(arr[1..n], n)

quickSort(arr[1..p-1], p-1)

quickSort(arr[p+1..n], n-p)

QuickSort [$O(n \log n)$ version] is non-stable.

Duplicates Partitioning: 3 way partitioning: Packing duplicates together!

One pass:

Two passes:

Heap: Largest j items are sorted in the final j positions after the j -th iteration.

void heapSort(arr, n)

for n - 1:

1. Swap root with last element

2. Bubble down the root node

Heapsort is non-stable.

4 Trees ADT

insert(Key k, Value v) delete(Key k)

search(Key k) contains(Key k)

successor(Key k) size()

predecessor(Key k)

Runtime: Most methods are $O(\text{height})$

K search(K curr, V value)

if (value == curr) then

return curr

else if (value == null) then

return null;

else if (value > curr.key) then

return search(curr.right, value)

// Symmetrical on the left

void insert(K curr, V value)

if (value > curr.key) then

if (curr.right == null) then

curr.right = value

else then

insert(curr.right, value)

// Symmetrical on the left

K successor(K curr, V value)

result = search(curr, value)

if (result > value) then return result;

else if (result <= value) then

Return the right child.

// If no right child and result is the

// left child of parent, return parent.

// Otherwise recurse until it is a left

// child.

void delete(K curr, V value)

result = search(curr, value)

if (result.hasNoChildren()) then

result.delete()

else if (result.hasOneChild()) then

result.parent.setChild(result.child)

else if (result.hasTwoChildren()) then

s = successor(result) // MAX 1 child

swap(s, result)

delete(result, result)

Balanced Property

BST is balanced if $h \in O(\log n)$

1. Define a "good property" of a tree

2. Prove that the "good property" $\Rightarrow$ balanced tree

Height balanced BST with height h has at least $n > 2^{h/2}$ nodes $\equiv n$ nodes has height $h < 2 \log n$

3. After every operation, reestablish property as invariant

AVL Tree: $|v.\text{left}.\text{height} - v.\text{right}.\text{height}| \leq 1$

BST is height balanced $\iff \forall v \in \text{BST}, v$ is height balanced

Proof: $n_h \geq 1 + n_{h-1} + n_{h-2} \Rightarrow n_h \geq 2^{h/2}$ Maintain balance: If v is left-heavy:

1. $v.\text{left}$ is balanced: rightRotate(v)

2. $v.\text{left}$ is left-heavy: rightRotate(v)

3. $v.\text{left}$ is right-heavy: leftRotate($v.\text{left}$) + right-rotate(v)

Note: Case 1 does not decrease root height. Only results from deletion, so no need to further decrease root height. Requires up to $O(\log n)$ rotations

(a, b)-Tree: $2 \leq a \leq (b+1)/2$

Node type	#Keys		#Children	
	Min	Max	Min	Max
Root	1	$b-1$	2	b
Internal	$a-1$	$b-1$	a	b
Leaf	$a-1$	$b-1$	0	0

A non-leaf node must have one more child than its number of keys. The keys specify the range each child falls under.

All leaf nodes must be at the same depth.

merge, share, split

Order Statistics Tree: AVL Tree, storing subtree weight

5 Hashing

insert(Key k, Value v) contains(Key k)

search(Key k) size()

delete(Key k)

Direct Access Table

Using an array indexed by keys to access values. Suppose English words, with max. 28 letters, and each letter represented in 5 bits. Any word can then be represented in 140 bits. We require a 2^{140} sized direct-access array.

Hash Functions $h: U \mapsto \{1..m\}$

Considers the smaller set of actual keys K , and map the $|K| = n$ keys to $m \approx n$ buckets. By PHP, $\exists h(k_1) = h(k_2) \wedge k_1 \neq k_2$.

Simple Uniform Hashing Assumption: Every key is equally likely to map to every bucket, and keys are mapped independently.

Let $X_i = \text{numBalls}$ in the i -th bucket. $X_i \sim \text{Bin}(n, \frac{1}{m})$

For fixed load α , $E(\text{size of max bucket}) = \Theta(\frac{\log n}{\log \log n})$

int Object::hashCode(): Must redefine equals. If two objects are equal, they have to return the same hash code.

Used: o.hashCode() ^ (table.size() - 1)

Dealing with Hash Collisions

• New Hash Function

"Better"? Need to copy table, and eventually another collision

• Chaining

• Open Addressing

Chaining: Each bucket contains a linked list

Space: $O(m \text{ buckets} + n \text{ items})$

Runtime: Insert $O(1 + \text{cost}(h))$ Delete $O(n + \text{cost}(h))$

Open Addressing: Probing a sequence of buckets until empty found

Expected $O(1 + \frac{\alpha}{2})$

Space: $O(m)$ **Runtime:** Worst case $O(n)$ Practice $O(1)$ for constant load factor and good hash functions

Table Resizing

1. Choose new table size m'
2. Choose new hash function h' such that $h : U \mapsto \{1..m\}, h' : U \mapsto \{1..m'\}$
3. $\forall x \in \text{currTable}, \text{newTable}[h'(x)] = x$

Resizing costs $\Theta(m' + m + n)$ for new table size m' , current table size m , n elements.

Proof: Consider sized $2n$ table of n elements, and a sequence of n operations. n inserts or deletions are both $\Theta(n)$.

Cuckoo Hashing

```
insert(Key k, Table T, Hash h)
    slot = h(k)
    displaced = T[slot]
    T[slot] = k
    if (displaced != null) then
        insert(displaced,
                T == A ? B : A,
                T == A ? g : f)
```

6 Binary Heap

```
insert(int pri, Key k) size(Key k)
extractMax() peekMaxId()
contains(Key k) peekMaxPriority()
increaseKey(int pri, Key k)
```

Consists: indices $\mapsto$ priorities, id $\mapsto$ indices, indices $\mapsto$ id

Invariants: Priority of each node < Priority of parent

Heap is a complete binary tree

Runtime: Mostly $O(\log n)$

insert: Set next = ele, while not root and >parent: bubbleUp

increase/decreaseKey: Change priority and bubble up-downwards

extractMax: Swap root with last element, remove new last element, bubble downwards

Building a Heap

For parent x of two subheaps L, R. Prove by strong induction that if we bubbleDown(x), the resulting heap satisfies invariants.

```
void heapify() {
    for (int idx = size/2; idx >= 1; idx--) {
        bubbleDown(idx);
    }
}
```

Heapify is $O(n)$: $1 \cdot \frac{n}{4} + 2 \cdot \frac{n}{8} + \dots + (h-1) = \sum_{k=1}^h \frac{kn}{2^{k+1}} = \frac{n}{4} \sum_{k=1}^h \frac{k}{2^{k-1}}$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

7 Graphs

Graph $G = (V, E)$, where $|V| > 0, E \subset \{(v, w) : (v \in V), (w \in V)\}$
Solving graph problems: Reduce problem to a graph, where each node is a state. Allows for stateless algorithms to run. If possible, use DAG (much more efficient) [Phantom nodes, 0-weight edges]

Graph Representations

If space is limited, $|E| \ll |V|^2$, use adjacency list.

If require matrix operations, use adjacency matrix.

Adjacency List $|V|$ -sized array, each element containing a linked list of neighbours

Space: $O(|V| + |E|)$, **Runtime:**

Iterating neighbours of specified v : $O(\deg(v))$

Determine if x, y are neighbours: $O(\min\{\deg(x), \deg(y)\})$

Adjacency Matrix $|V|^2$ -sized symmetric matrix, $A[v][w] = 1 \iff (v, w) \in E$. A^2 to find out 2-hop neighbours.

Space: $O(|V|^2)$, **Runtime:**

Iterating neighbours of specified v : $O(|V|)$

Determine if x, y are neighbours: $O(1)$

Edge List $|E|$ -sized array of all edges in the graph

Space: $O(|E|)$, **Runtime:**

Iterating neighbours of specified v : $O(|E|)$

Determine if x, y are neighbours: $O(|E|)$

Searching

Input: Source vertex S

Output: Visit destination vertex D . OR all nodes in the graph.

```
boolean bfs(source, dest)
    source.isVisited = true
    que.offer(source)
    while (!que.isEmpty())
        curr = que.poll()
        for (nbr : curr.nbrList())
            if (!nbr.isVisited) then
                nbr.isVisited = true
                que.offer(nbr)
                // Set nbr.dist/nbr.parent
    return dest.isVisited
```

DFS is exactly the same with a stack

BFS & DFS is $O(|V| + |E|)$

Topological Sorting

Ordering of nodes: $(u, v) \in E \Rightarrow u$ before v in the toposort.

Post-order DFS: **Runtime:** $O(|V| + |E|)$

```
void topo(Node[] nodes)
    for (node : nodes)
        if (!node.visited) then
            node.visited = true
            toposort(node)
            order.prepend(node)
void toposort(Node node)
    for (nbr : node.nbrList())
        if (!nbr.visited) then
            nbr.visited = true
            toposort(node)
            order.prepend(nbr)
```

Tarjan's Algorithm

To find cycles, points which removed disconnects graph (articulation points), strongly connected components ($\forall (u, v) \in V^2, \exists \text{path}(u, v)$)

$\text{stk} = \text{new Stack}() ;$

```
void tarjan(Node curr, int time)
    stk.push(curr)
    curr.time = time
    lowTime = curr.time
    for (Node nbr : curr.nbrList())
        if (nbr.lowTime is set) then continue
        if (nbr.time is set) then
            lowTime = min(nbr.time, lowTime)
        else if (nbr.time is not set) then
            tarjan(nbr, time + 1)
            lowTime = min(nbr.lowTime, lowTime)
    curr.lowTime = lowTime
    if (lowTime == time) then
        while (stack.peek() != curr)
            stack.pop() // SCC rooted at curr
    S is an articulation point if
```

- S is the source and in DFS tree, S has outdegree 2
- S is not the source, and has a neighbour v st. $v.\text{lowTime} \geq S.\text{time} \vee S.\text{lowTime} \geq v.\text{time}$

$\exists v \in V (v.\text{lowTime} < v.\text{time}) \Rightarrow \exists$ a cycle

Dijkstra

Triangle Inequality for (shortest) distance between Nodes:

$$\delta(S, C) \leq \delta(S, A) + \delta(A, C)$$

```
void dijkstra()
    for (node : nodes)
        pq.insert(node, inf)
    pq.decreaseKey(source, 0)
    while (!pq.isEmpty())
        curr, distance = pq.extractMin()
        dist[curr] = distance
        for (nbr : curr.nbrList())
            if (pq.contains(nbr))
                relax(nbr,
                    distance + E[curr, nbr])
```

Runtime: $O(|V| \log |V| + |E| \log |V|)$

Problem Formulation: To ensure that estimates monotonically decrease with relaxation with more nodes, and do not decrease further after it is popped from the priority queue. $\Rightarrow$ No negative edges

Bellman Ford

```
void bf()
    // Set up int[] dist
    for i = 1..|V|-1:
        for edge (u, v) in E:
            relax(dist, u, v)
        for edge (u, v) in E: // +1
            relax(dist, u, v)
    // If no change, no negative cycles
```

Runtime: $O(|V||E|)$ **Special Case-DAG:** $P_0 = 0, P_i = \{s_i\}, P_i = \{s_i\}$

```
// Set up int[] dist
// Get toposorted nodes topoList
for u in topoList:
    for neighbour v of u:
        relax(dist, u, v)
```

8 MST

Properties of MST

Suppose a connected graph. Otherwise, no spanning tree exists.

- MST is a acyclic graph. Tree.

- If an edge is removed from a MST, the two components are MSTs.
- For every cycle, the maximum weighted edge is not in the MST.
- For every cut of the nodes, the minimum weighted edge across the cut is in the MST.

Edge colouring for proof of correctness

Red Rule: If C is a cycle with no red arcs, then the max-weighted edge in C is coloured red. **Blue Rule:** If D is a cut with no blue arcs, then the min-weighted edge in D is coloured blue.

UFDS

Weighted Union: To make the smaller tree the subtree of the larger tree $\Rightarrow O(\log(n))$ height. **Runtime:** $O(\log n)$

Path Compression: Shrinks the tree whenever find is called.

Runtime: m operations on n objects- $O(n + m\alpha(m, n))$

```
findRoot(int p)
    root = p;
    while (parent[root] != root)
        root = parent[root]
    while (parent[p] != p)
        temp = parent[p];
        parent[p] = root;
        p = temp;
    return root;
```

MST for MINI MAX!

Kruskal's Algorithm

Runtime: $O(E\alpha(E)) + O(E \log E) = O(E \log V)$

- Initialise UFDS for n nodes
- Sort edges by weights
- For each edge $e = (u, v)$:
 - If u, v are in the same component, skip
 - Otherwise, add e , and union u, v

Prim's Algorithm

Runtime: $O((E + V) \log V)$. $O(E + V \log V)$ for fibo-heap.

Basic idea:

- S : set of nodes connected by blue edges
- Initially, $S = \{A\}$
- Repeat:
 - Identify cut: $\{S, V \setminus S\}$
 - Find minimum weighted edge on cut
 - Add new node to S

Variants

Undirected graphs / Equal weighted edges: DFS / BFS!

DAGs: $\forall v \in V$ add minimum weighted incoming edge- $O(E)$

Edges weighted $\{1..10\} \Rightarrow$ use an array of size 10 to act as PQ

Reweightings: Only relative edge weights matter. Addition, Multiplication allowed. To find MaxST, multiply by -1 and run MST.

9 Dynamic Programming

Everything is a table!

Optimal Sub-Structure

Optimal solution can be constructed from optimal solutions to smaller subproblems

Doesn't always exist If $P = (u \rightarrow v \rightarrow w)$ is the shortest path from u to w , P contains shortest path from u to v & v to w .

DP Recipe

- Identify optimal substructure
- Define subproblems
- Solve problem using subproblems
- Write pseudocode

DP Analysis Let $S[v, w, P]$ be shortest path from v to w using nodes from P . Base case $S[v, w, \emptyset] = \infty$

- Count subproblems
- Figure out total time to solve all subproblems